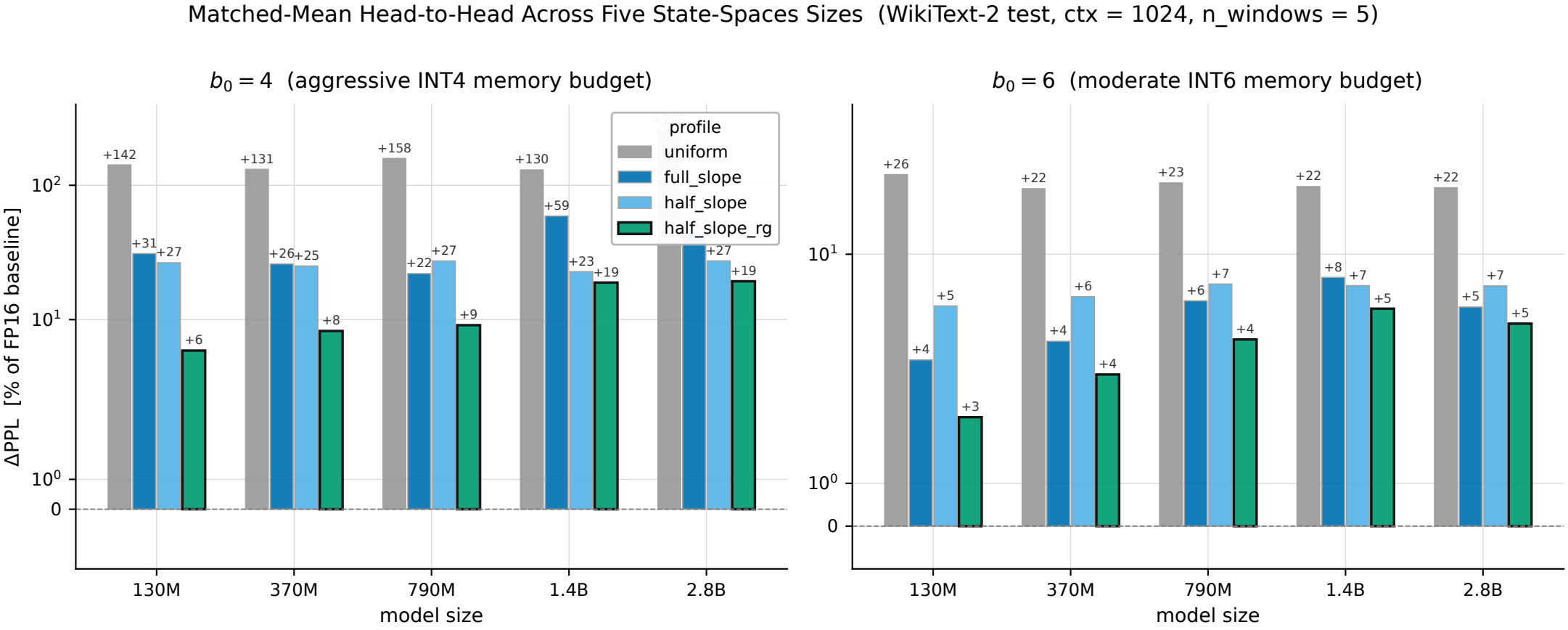
Matched-Mean Head-to-Head Across Five State-Spaces Sizes (WikiText-2 test, ctx = 1024, n_windows = 5)
$b_0 = 4$ (aggressive INT4 memory budget)
$b_0 = 6$ (moderate INT6 memory budget)
ΔPPL [% of FP16 baseline]
model size
130M
370M
790M
1.4B
2.8B
profile
uniform
full_slope
half_slope
half_slope_rg
+142
+31
+27
+6
+131
+26
+25
+8
+158
+22
+27
+9
+130
+59
+23
+19
+27
+27
+19
+26
+4
+5
+3
+22
+4
+6
+4
+23
+6
+7
+4
+22
+8
+7
+5
+22
+5
+7
+5